

Shashank Dhiman

Full Stack Developer | Back-end & Systems Engineering

+91 88476 80989 | dhimanshashank15@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode](#)

PROFESSIONAL SUMMARY

Full Stack Developer with 1+ year of production experience building real-time communication systems, AI-integrated video proctoring, and event-driven analytic pipelines. Proficient in Node.js, PostgreSQL, Redis, WebRTC, AWS Lambda, and ClickHouse. Delivered measurable results across multiple production systems — 70% API latency reduction, 200+ concurrent proctoring streams, and sub-second analytic on thousands of daily LMS events — in an Agile product environment.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Java, Python, SQL

Front-end: React.js, Redux Toolkit, Tailwind CSS, Material UI

Back-end: Node.js, Express.js, REST APIs, Socket.IO, Microservices, AWS Lambda, Serverless

Databases: PostgreSQL, ClickHouse, Redis, MongoDB

Real-Time & AI/ML: WebRTC (Media-soup), Media Pipe, Event-Driven Architecture

DevOps & Tools: Docker, Git, Linux, Postman, CI/CD

EXPERIENCE

Venture Pact – Masters' Union

January 2025 – Present

Associate Software Developer

Phagwara, India

- **Built a WebRTC proctoring platform** (Media-soup) supporting **200+ concurrent student streams** with live admin dashboards and real-time state sync; integrated a front-end proctoring micro-service (Media Pipe) for automated eye-gaze tracking, tab-switch detection, and behavioral classification during online assessments.
- Refactored the proctoring pipeline with async flow management and connection pooling, boosting system throughput by **~35%** under peak load; optimized PostgreSQL indexing, reducing proctoring event query time by **~50%**.
- **Engineered the real-time Messaging module end-to-end** — DM and group chat with read receipts, and Redis-backed presence tracking; replaced offset pagination and a Redis caching layer, cutting API latency by **70%** and database load by **~40%**.
- Eliminated N+1 query patterns through batched querying, reducing database round-trips by **~60%** for high-concurrency chat-history retrieval.
- **Architect an event-driven analytic pipeline** (AWS Lambda + Click-House) ingesting thousands of LMS activity events daily; implemented to YYYYMM partitioning and composite indexing, improving analytic query performance by **~55%**.
- Designed a regex-based event classification engine mapping raw events into **9+ dashboard categories** with multi-tenant filtering across 6 dimensions (org, cohort, program, session, date, user), enabling zero-code taxonomy updates for new event types.
- Contributed to feature design, code reviews, and production releases across **3+ parallel product tracks** in a fast-paced Agile (Scrum) environment.

Qspider

June 2024 – July 2024

Software Developer Intern

Chandigarh, India

- **Built Eventify**, a full-stack event management platform (MERN) with Stripe payment integration, JWT authentication, and Role-Based Access Control (RBAC) — handling secure transactions and scoped user permissions end-to-end.
- Designed a modular RESTful API architecture with middleware chaining and error-boundary patterns, achieving clean separation of concerns and straightforward feature extension.

EDUCATION

Guru Nanak Dev Engineering College, Ludhiana

August 2021 – June 2025

Bachelor of Technology – Computer Science & Engineering

CGPA: 8.45 / 10

Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, System Design